

Introduction to Multilevel Measurement Models

SMIP Summer School 2025: Lecture 01

About Your Instructors

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Workshop Introduction

This is an ambitious workshop:

- Measurement models (Item Response Theory)
- Multilevel models for clustered data
- Bayesian statistics
- Multilevel measurement models

Each could be its own workshop

Workshop Materials

Set of slides on topics pulled from other course materials

- In case additional detail on any topic is needed

Full set of analysis files in R and Stan

- In case you would like to work ahead

Workshop Schedule

- Mornings: 9:00-12:30 (**Including Friday**)
 - Break at 10:45
- Afternoons:
 - Tuesday: 14:00-17:15 (break at 16:00)
 - Wednesday: 15:15-17:45 (break at 16:45)
 - Thursday: 14:00-15:45 (no break)

Lecture and Example Time Plan

Last hour of each day is open

Your choice:

- Questions
- Example practice
- Work with your data
- Leave early

Syntax and Model Note

- As stan models can take a very long time to run, all prior model results have been saved
- All lectures are in self-contained folders
- All files are available at <https://github.com/jonathantemplin/MultilevelMeasurementModels2025>

Workshop Sections

1. Introduction to Measurement Models, Bayesian Statistics, and Stan
2. Introduction to Multilevel Models
3. Introduction to Multilevel Measurement Models

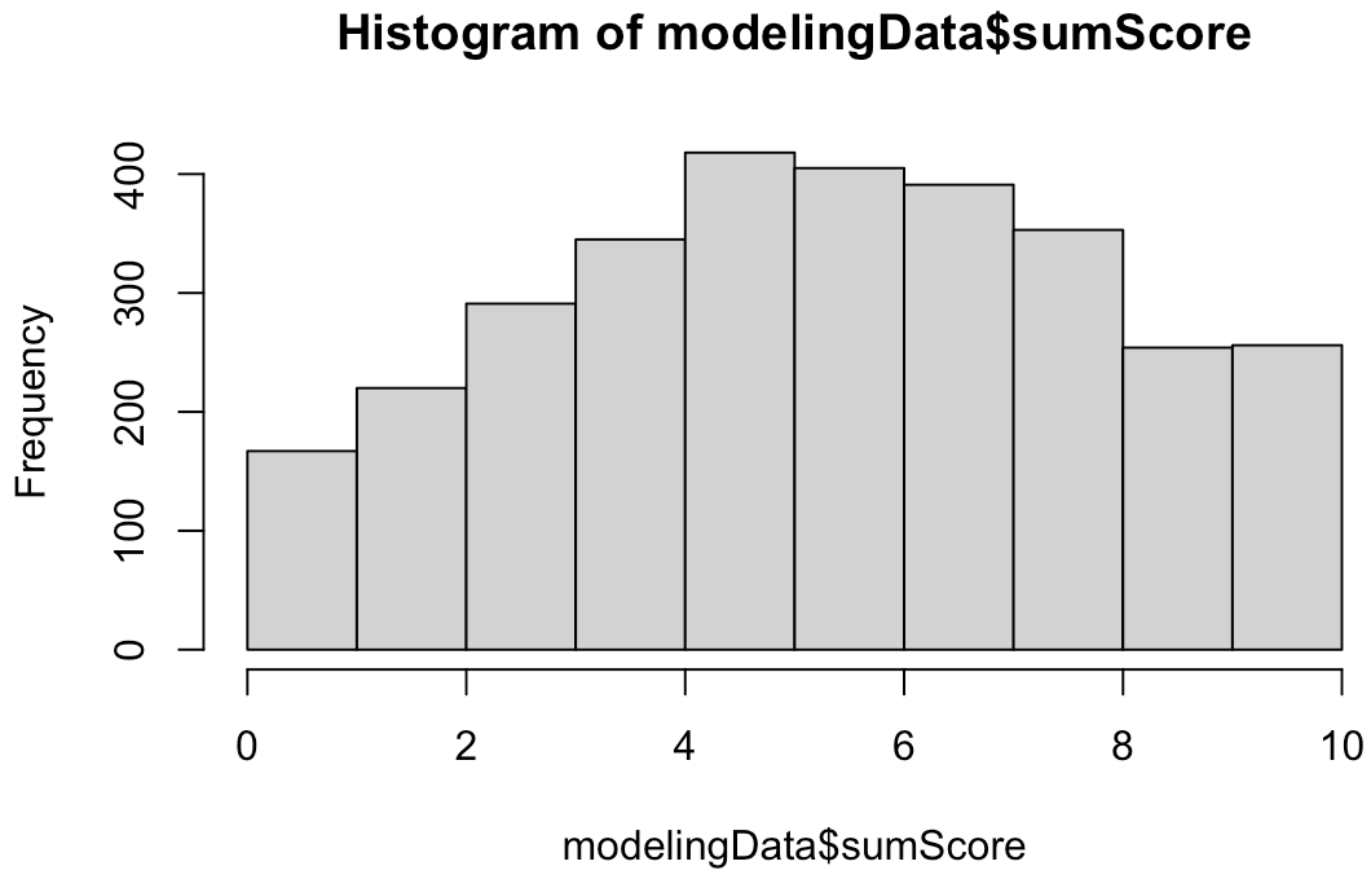
Running Example

Running Example Data

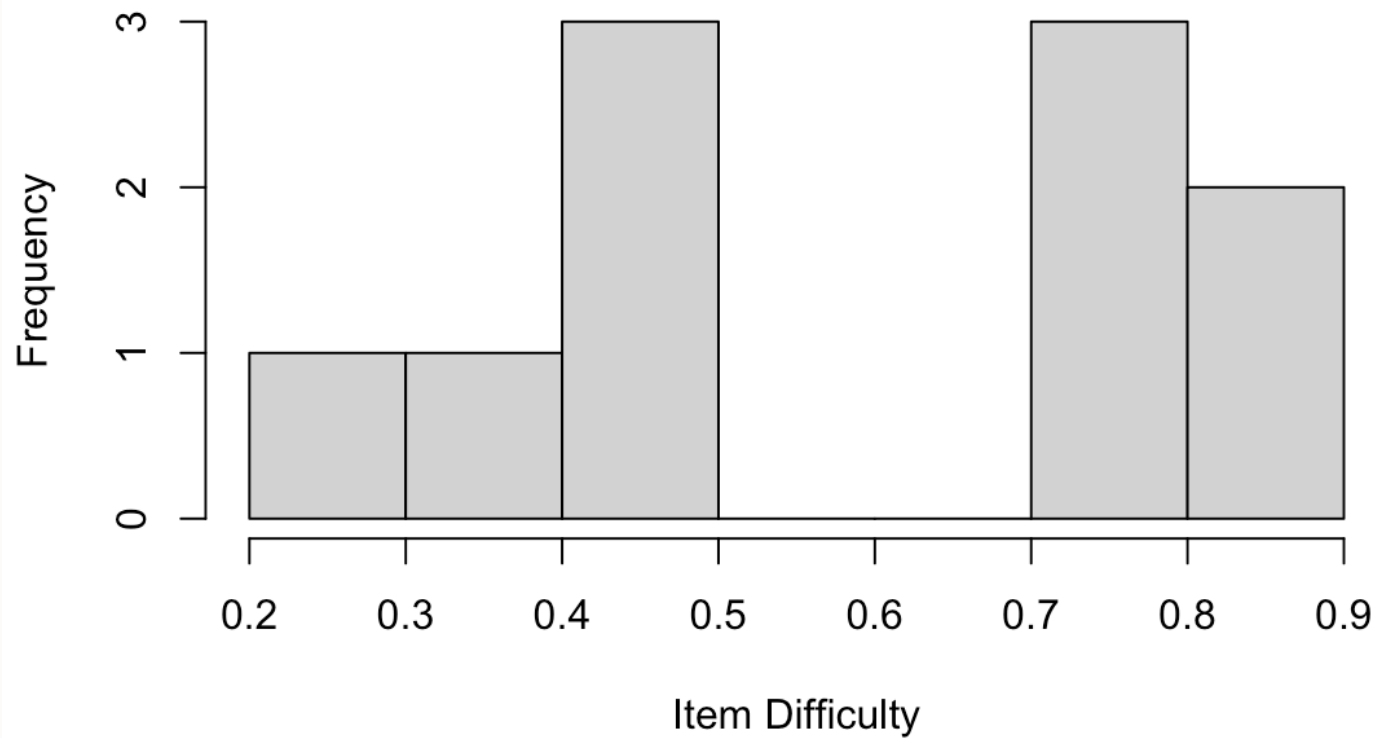
Data come from a secondary school (10th grade) end-of-grade mathematics assessment given around the year 2006 in a midwestern state in the USA

- Sample of 50 students from 62 schools
- Sample of 10 mathematics items from the assessment
 - Items are scored correct/incorrect (`score1` - `score10`)
 - The sum of all 10 items is included (`sumScore`)
- Other variables:
 - `studentID` : Student ID number (created for this example)
 - `schoolID` : School ID number (created for this example)
 - `frlunch` : Free/reduced lunch status (1 = free/reduced lunch, 0 = not free/reduced lunch; indirect indicator of student socioeconomic status)

Distribution of Sum Scores



Distribution of Item Difficulties



Workshop Lectures

- Lecture 01: Introduction to Multilevel Measurement Models Workshop
- Lecture 02: Introduction to Measurement Models
- Lecture 03: Introduction to Bayesian Statistics
- Lecture 04: Introduction to MCMC and Stan
- Lecture 05: Bayesian IRT Models in Stan
- Lecture 06: Introduction to Multilevel Models
- Lecture 07: Bayesian Multilevel Models in Stan
- Lecture 08: Bayesian Multilevel Measurement Models in Stan